

Math 1314 Lesson 24

1)  $f(x) = \frac{1}{x+4}$ .

a) Graph  $f(x) = \frac{1}{x+4}$

b)  $f(-4) = \underline{\hspace{1cm}}?$

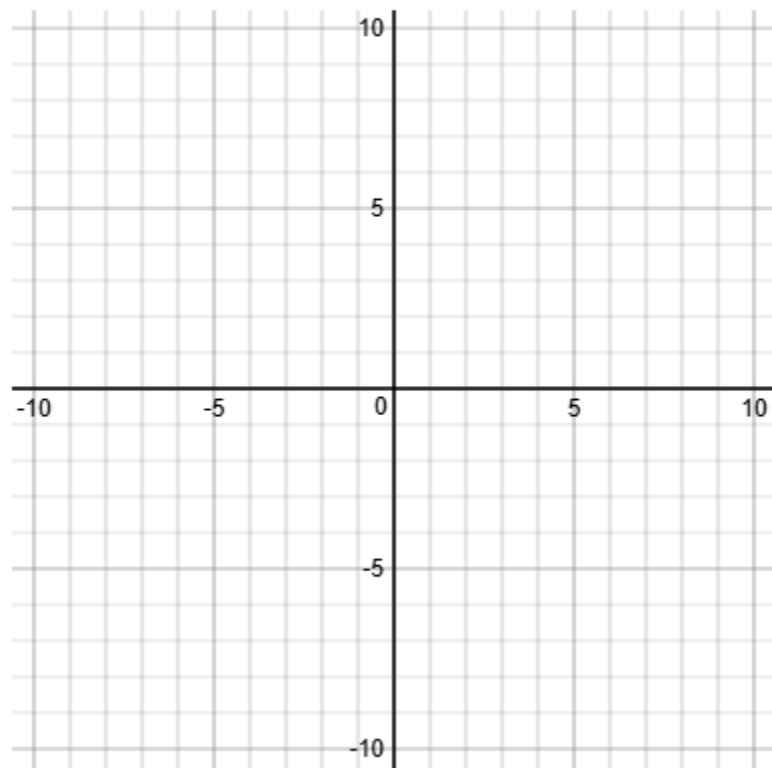
c) Graph splits up at  $x = \underline{\hspace{1cm}}?$

d) How far to the left does graph extend?

e) How far to the right does graph extend?

f) Domain of  $f(x) = \frac{1}{x+4}$  is  $\underline{\hspace{1cm}}?$

g) Vertical Asymptote is  $\underline{\hspace{1cm}}?$



2)  $f(x) = \frac{-4}{x-5}$ .

a) Graph  $f(x) = \frac{-4}{x-5}$

b)  $f(5) = \underline{\hspace{1cm}}?$

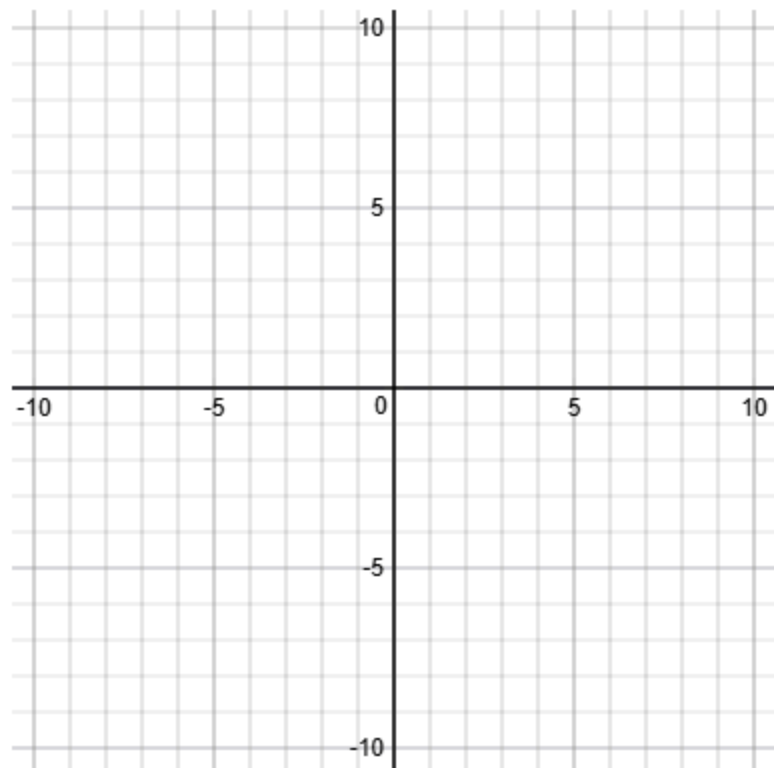
c) Graph splits up at  $x = \underline{\hspace{1cm}}?$

d) How far to the left does graph extend?

e) How far to the right does graph extend?

f) Domain of  $f(x) = \frac{-4}{x-5}$  is  $\underline{\hspace{1cm}}?$

g) Vertical Asymptote is  $\underline{\hspace{1cm}}?$



3) Find the vertical asymptote of the function  $f(x) = \frac{-2}{3x-6}$ .

a) Graph  $f(x) = \frac{-2}{3x-6}$

b)  $f(\underline{\quad?}) = \text{undefined}$

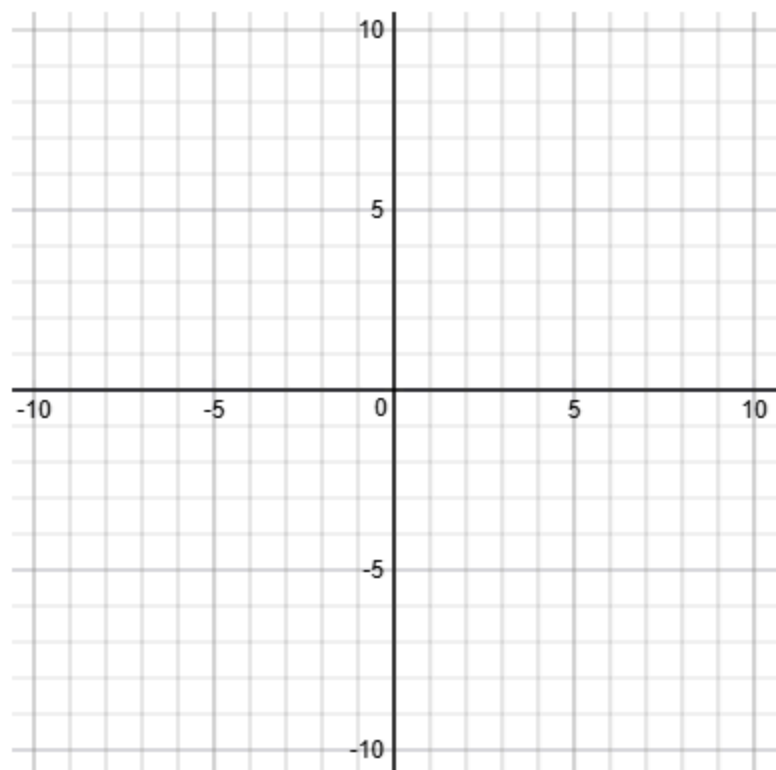
c) Graph splits up at  $x = \underline{\quad?}$

d) How far to the left does graph extend?

e) How far to the right does graph extend?

f) Domain of  $f(x) = \frac{-2}{3x-6}$  is  $\underline{\quad?}$

g) Vertical Asymptote is  $\underline{\quad?}$



4) Find the vertical asymptote of the function  $f(x) = \frac{x}{x^2 - 16}$ .

a) Graph  $f(x) = \frac{x}{x^2 - 16}$

b)  $f(\underline{\quad?}) = \text{undefined}$  (there are two  $x$ -values)

c) Graph splits up at  $x = \underline{\quad?}$

d) How far to the left does graph extend?

e) How far to the right does graph extend?

f) Domain of  $f(x) = \frac{x}{x^2 - 16}$  is  $\underline{\quad?}$

g) Vertical Asymptotes are  $x = \underline{\quad?}$  and  $x = \underline{\quad?}$

